PAPER_720: Doc 43: Universal Permanence Equation, AGN Feedback, and the Final Parsec Problem

Class: `UQFFKnowledgeBaseKB5` **CP4 Entry:** #304 **Keywords:** UQFF, Universal Permanence, AGN feedback, Final Parsec, SMBH binary, non-local jump, metal retention, f_Z
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Abstract

Doc 43 (UFE ORB EXP 2_24_07Mar2025) presents the Universal Permanence (UP) equation, a master UQFF sum including all field contributions (U_{gi}, U_m, TRZ zone, QS factor), applied to three astronomical phenomena: the Red Dwarf plasma orb (Drawing 30, Bearden TRZ), AGN feedback mass-metallicity relation (Drawing 31, Sanchez et al.), and the Final Parsec problem for SMBH binaries. The non-local jump probability $P \approx 0.49$/s demonstrates emergent quantum jumps in the UQFF framework.

1. Universal Permanence Equation

$$\mathrm{UP}(t) = \sum_i k_i U_{gi} + \sum_j \frac{\mu_j}{r_j}(1 - e^{-\gamma t^-} \cos(\pi t_n))\hat{\phi}_j U_{mj} + T_s^{\mu\nu} + U_b + NN + QS + ACE + DCE + [SSq] +$$

$$\approx 1.03 \times 10^{20} \cdot QS$$

where:

$$t^- = -t_n \cdot e^{\pi - t_n} \qquad (t_n = 13.68\,\mathrm{s} \implies t^- \approx -3.75 \times 10^{-4}\,\mathrm{s})$$

2. Non-Local Jump Probability

$$P = 1 - e^{-\gamma |t^-|} \approx 0.490\,\mathrm{s}^{-1}$$

with $\gamma = 10^3$ s^{-1}. Observed: 1-1.5 non-local jumps per video frame.

3. Energy Density

$$\rho_{react}(t) = 10^{15} \cdot e^{-0.001 \cdot t_n} \approx 9.86 \times 10^{14}\,\mathrm{W/m^3}$$

4. Metal Retention Factor f_Z

Galaxy Type	f_Z
Over-massive SMBH	0.89
Under-massive SMBH	0.85
Star-forming disk	0.73
Quenched disk	0.51

$$f_{CGM} = \frac{M_{CGM}}{M_{vir}}$$

5. Final Parsec

$$U_{g4}^{FP}(t) = k_4 \cdot \rho_{SCm} \cdot \frac{M_{BH}^{binary}}{d_{parsec}} \cdot e^{-\alpha t} \cdot \cos(\pi t_n)$$

$M_{BH}^{binary} = 8.15 \times 10^{36}$ kg, $d_{parsec} = 2.55 \times 10^{20}$ m:

$$\boxed{U_{g4}^{FP}(0) \approx 7.64 \times 10^{-6}\,\text{J/m}^3}$$

6. AGN Feedback

$$U_{g4}^{AGN}(t) \approx 8.40 \times 10^{-6} \cdot e^{-0.001t} \cdot \cos(\pi t_n)\,\text{J/m}^3$$

for $\Delta M_{BH} = 1$ dex, $f_{feedback} \approx 0.1$.

References

- UQFF Doc 43 (UFE_ORB_EXP_2_24_07Mar2025.docx)
- grok_share_ba508f76c8e.txt entry #69
- Sanchez et al. AGN feedback mass-metallicity relation
- Session 176, v5.33

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